PrintNightmare refers to multiple vulnerabilities in the Windows Print Spooler service, most notably:

- CVE-2021-34527 – Remote Code Execution
- CVE-2021-1675 – Privilege Escalation

The flaw allows an attacker to load a malicious DLL via the Print Spooler service, leading to SYSTEM-level compromise.

Key Differences:

- **CVE-2021-1675 (Initial Vulnerability):** Originally identified as a Local Privilege Escalation (LPE) vulnerability, it allows an attacker with low-level access to the system to escalate privileges to SYSTEM level. It was patched in June 2021.
- **CVE-2021-34527 (The "True" PrintNightmare):** This is a critical Remote Code Execution (RCE) vulnerability that allows remote, unauthenticated attackers to execute arbitrary code with SYSTEM privileges. It was discovered after the initial patch for CVE-2021-1675 was released and found to be insufficient to prevent remote exploitation.

- # Network IOC Summary

Indicator Type	What to Look For
SMB Traffic	TCP 445 to suspicious internal host
Print Spooler	Unexpected remote driver install
File Creation	DLL in spool\drivers directory
Outbound Traffic	C2 beaconing after spooler crash
Process	`spoolsv.exe` spawning cmd.exe or powershell

Indicators of Compromise (IOC)

Windows Event Logs

Event ID 316 (PrintService Operational)

Event ID 808 / 812

Driver installation attempts

Event ID 4624

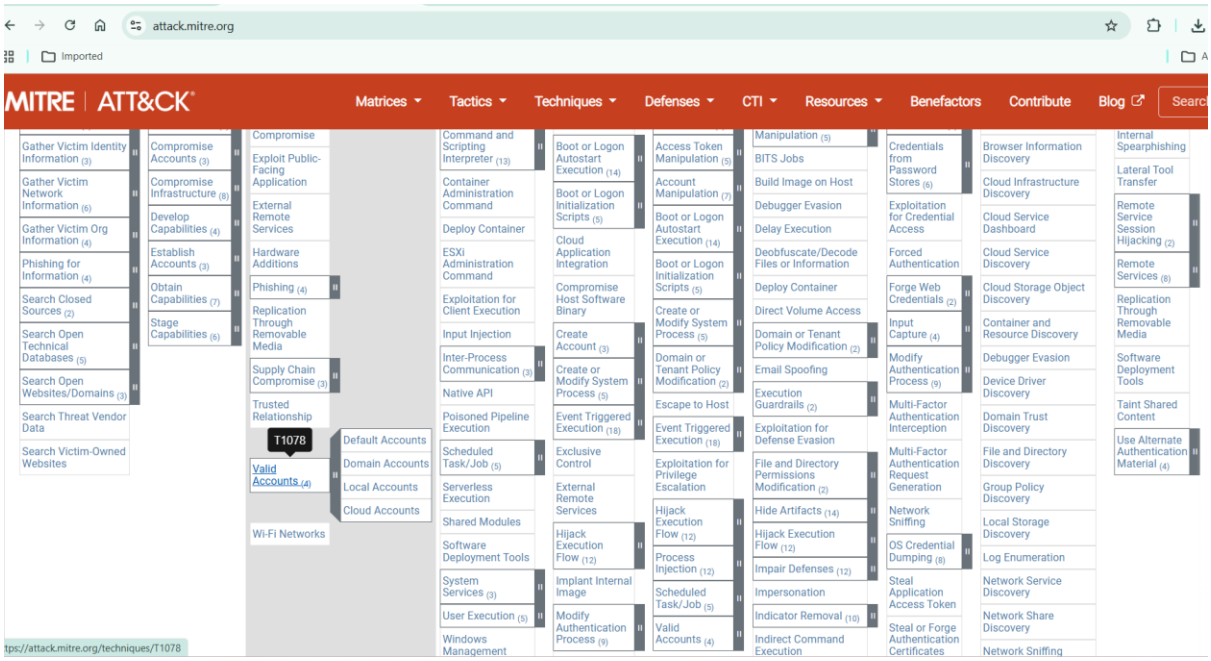
Logon Type 3 (Network logon)

Print Nightmare TTP

1) Initial access

T1078 – Valid Accounts

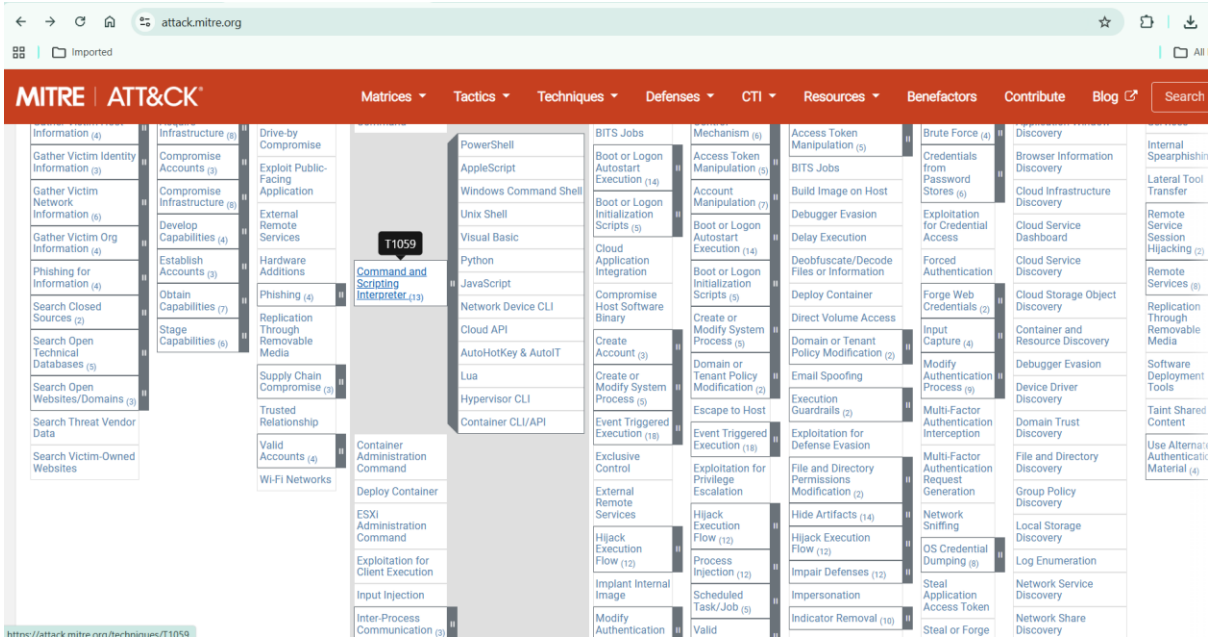
Domain user to reach RPC from print spooler service



2) Execution

T1059 – Command and Scripting Interpreter

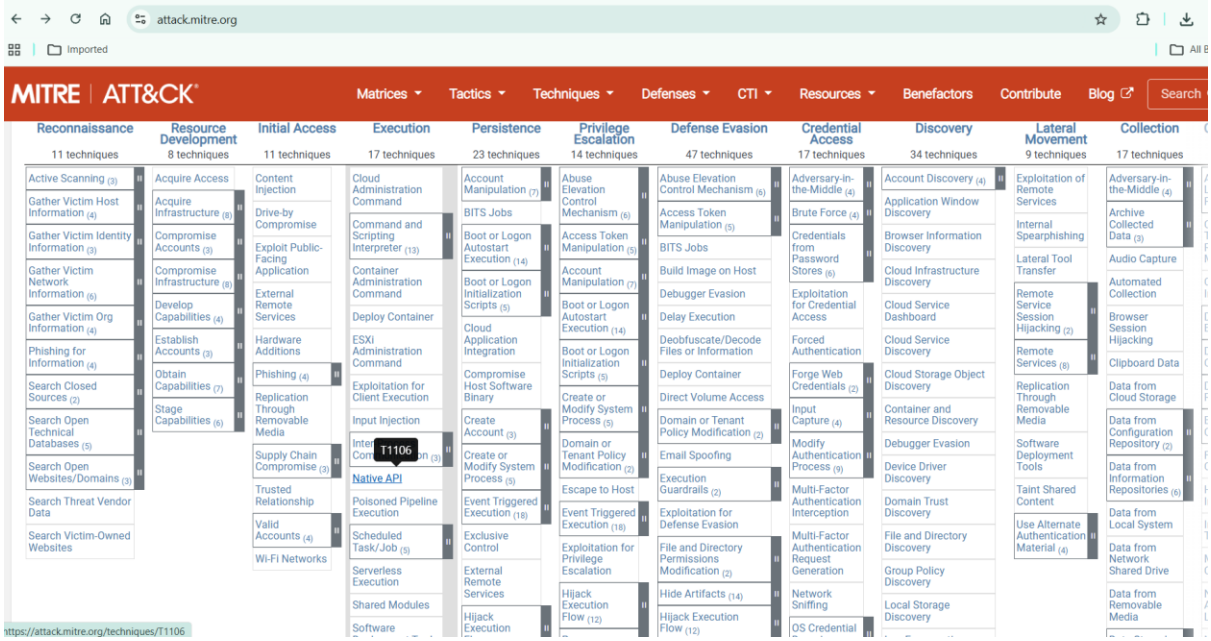
Uploading DLL by PowerShell



T1106 – Native API

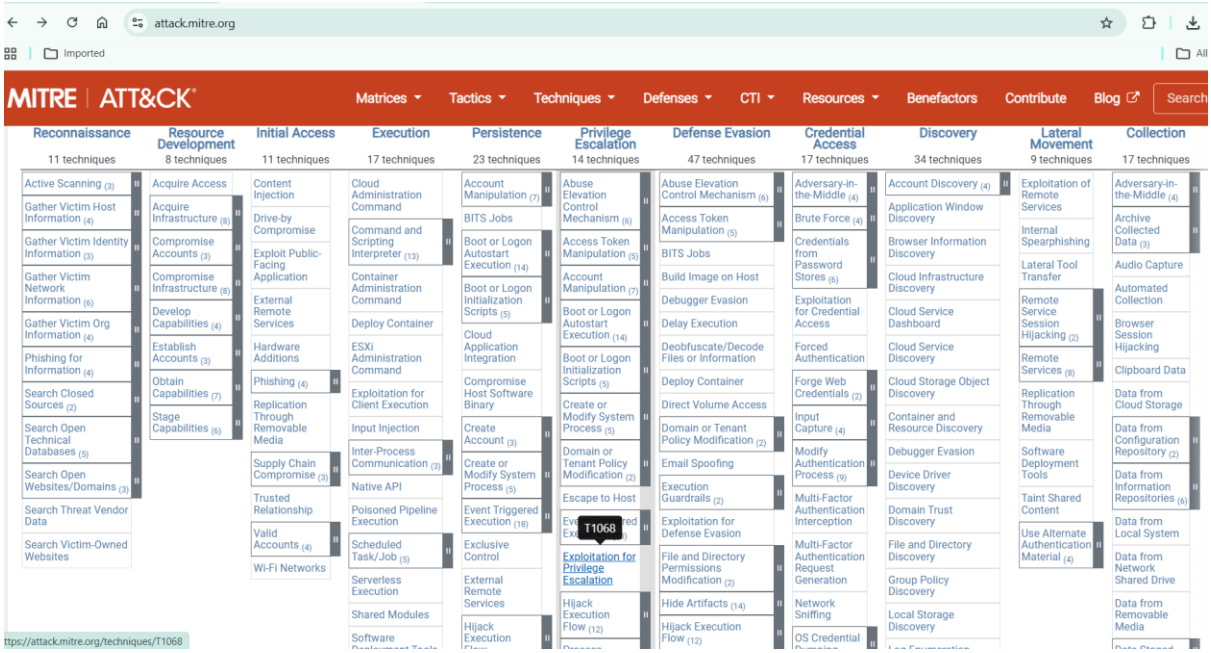
Using windows API

RpcAddPrinterDriverEx



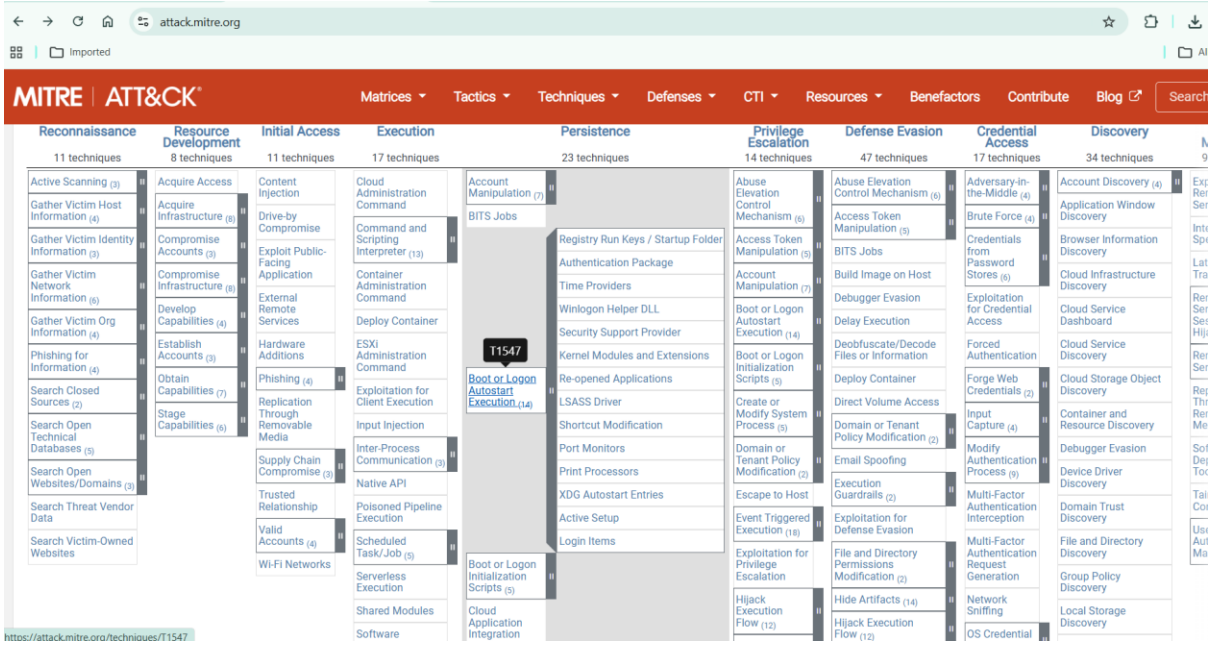
3) Privilege escalation

T1068 – Exploitation for Privilege Escalation

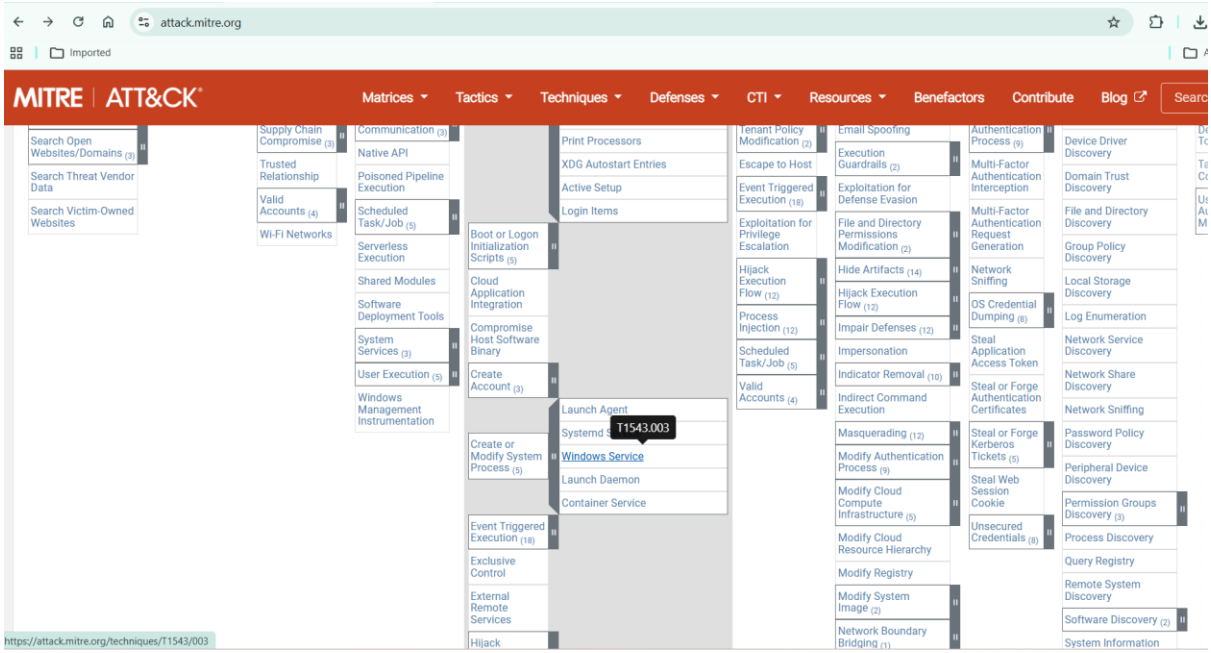


4) Persistence

T1547 – Boot or Logon AutoStart Execution



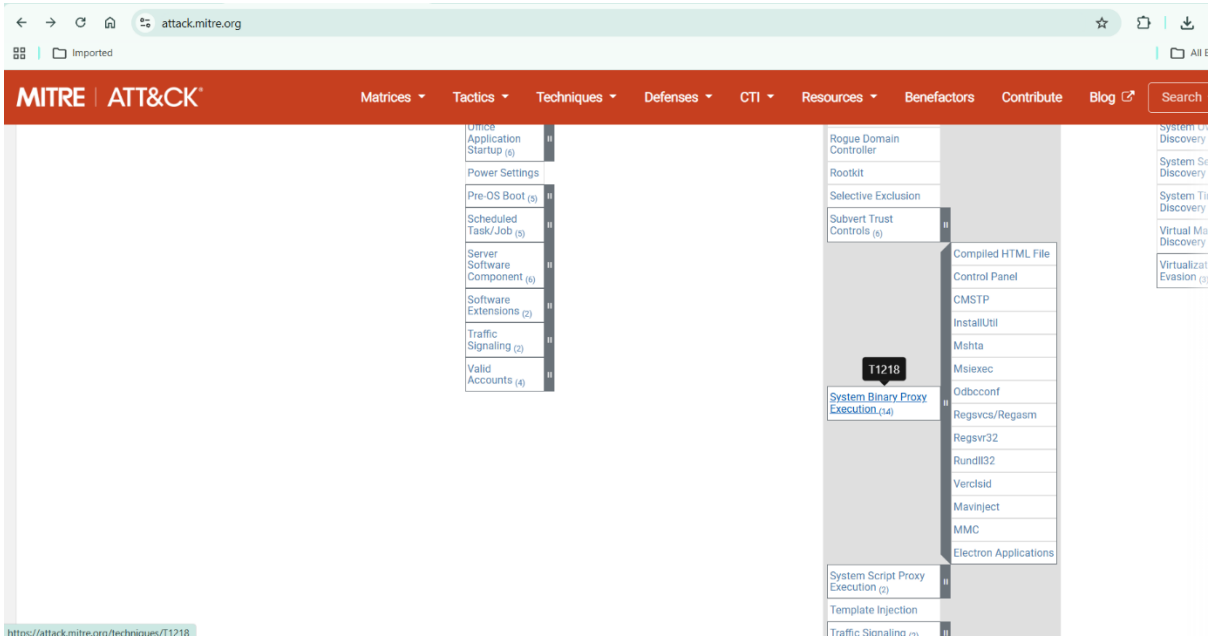
T1543.003 – Create or Modify System Process (Windows Service)



5) Defense evasion

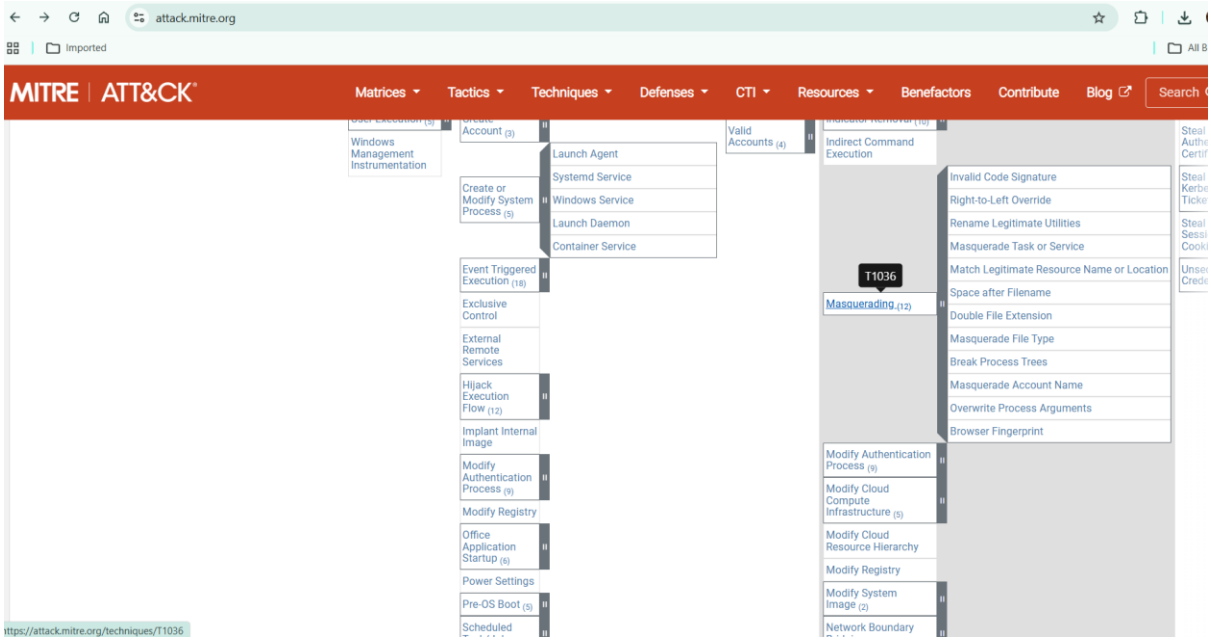
T1218 – Signed Binary Proxy Execution

spoolsv.exe



T1036 – Masquerading

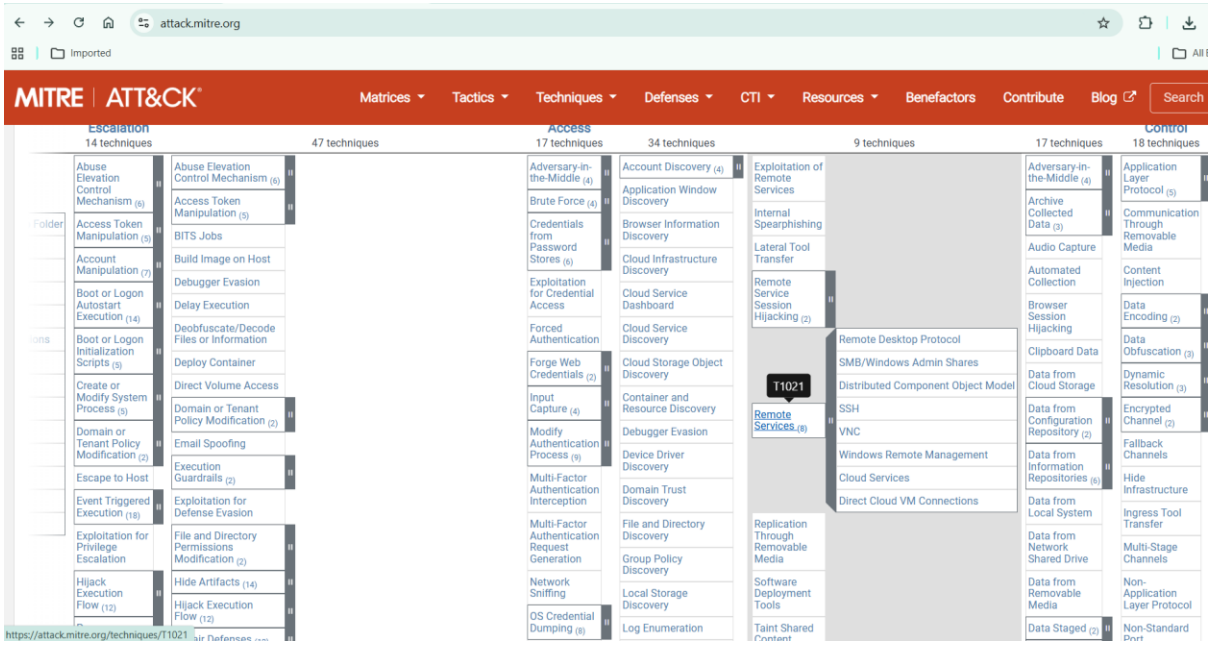
DLL as printer name



6) Lateral movment

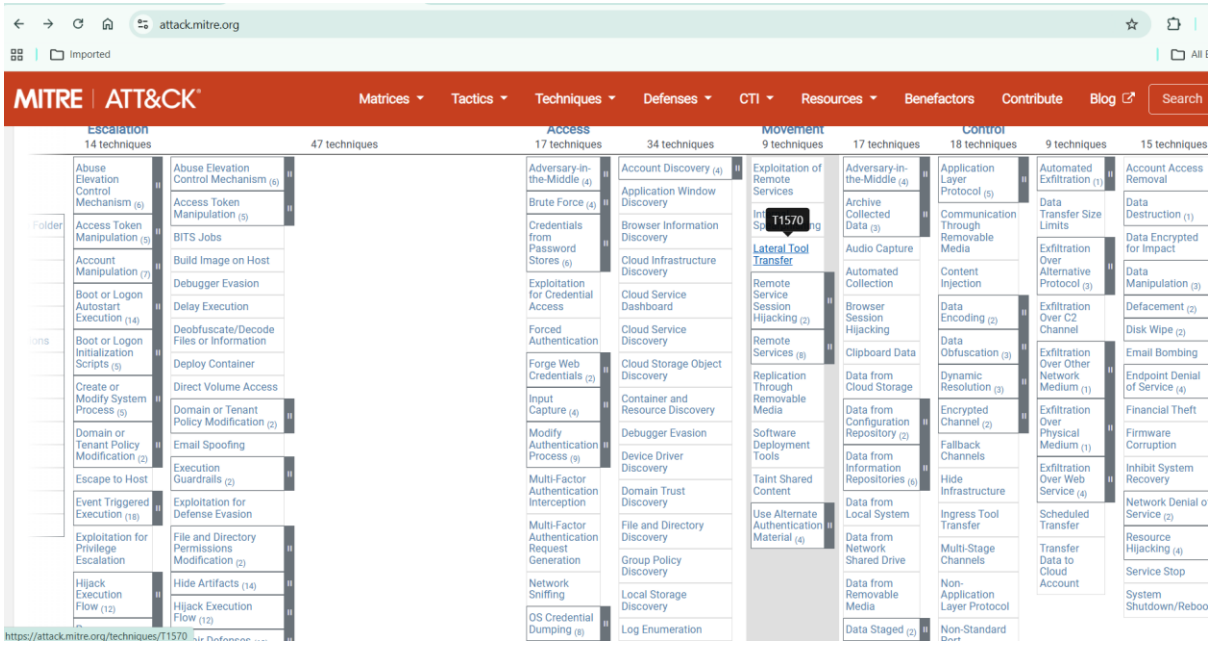
T1021 – Remote Services

Reach more devices by RPC/SMB



T1570 – Lateral Tool Transfer

Uploading DLL by SMB



7) Impact

T1484 – Domain Policy Modification

Control domain controller